Bibliography — Finnish Volleyball Player Development Model

Consolidated, deduplicated reference list for `finnish-volleyball-development-model-DRAFT.md`, its companion `finnish-volleyball-practice-design-guide-DRAFT.md`, and the research log `finnish-volleyball-model-research-notes.md`. This is the **single source of truth** for citations; the draft's §13 points here.

Conventions. APA-ish author–year, with a one-line "why it matters" note. Tags mark source type so peer-reviewed work isn't confused with practitioner/grey material:

- `[peer-reviewed]` journal article · `[book]` · `[thesis]` · `[review]` systematic/narrative review · `[grey]` practitioner blog/podcast/working paper · `[org]` federation/governing-body document.

Honesty flags. `[check: …]` marks metadata I could not fully verify and that should be confirmed before this goes anywhere adoption-facing.

Coverage note. This was built in two passes: (1) consolidating + verifying everything already cited across the drafts; (2) a deliberate gap-filling pass adding the foundational/adjacent literature the argument leans on but hadn't cited (eco-D foundations; the traditional motor-learning counterpoint; game-based pedagogies; the talent-development critique; perceptual-cognitive expertise; SDT and the Finnish coaching-competence model). It is now broad; it is not a formal systematic review.

---

1. Finnish volleyball federation & youth competition `[org]`

- **Suomen Lentopalloliitto.** *Lentopallo-Lajilinjaukset* (29 pp.) and *Lapsuusvaiheen-Lajilinjaukset* (11 pp.). — The federation's skill guidelines; primary source for the §1 "technique-first tilt" critique (*Periaatteet + Opetusavaimet*; "Yksinkertaiset liikkeet…"). Local PDFs; quotes in research-notes §3.
- **Suomen Lentopalloliitto.** *Mini-ikäisten (U9–U13) säännöt ja pelimuodot.* https://www.lentopallo.fi/valmennus/mini-ikaisten-valmennus/mini-ikaisten-saannot-ja-pelimuodot/ — Official statement of the youth game formats and the 2025–26 changes.
- **Junnulentis (Suomen Lentopalloliitto).** *Minisäännöt 2025–2026.* https://junnulentis.fi/2025/08/14/minisaannot-2025-2026/ — Full mini rules: U9 MiniLentis 2v2, U11/U13 3v3, Triplapallo (Triple Ball), free reception, Tiikeri/Super tiers, net heights, the *koppi* (catch) rule. Distilled in research-notes §9.
- **PowerCup.** *Säännöt* and *Sarjat.* https://www.powercup.fi/turnaus/saannot/ · https://www.powercup.fi/sarjat/ — The mass participation tournament; Tsemppi/Super series mirror the Tiikeri/Super tiers.
- **Suomen Lentopalloliitto.** Coach-education hub. https://www.lentopallo.fi/koulutukset/ — Landing page for the tutkinto path.
- **Suomen Lentopalloliitto.** *LTTV — Lentopallon tekniikka- ja taitovalmennus.* https://www.lentopallo.fi/koulutukset/tekniikkakoulu/ — Technique/skill foundation course (the one the model proposes to reframe).
- **Suomen Lentopalloliitto.** *Lasten lentopallovalmennus (LLV, U9–U13).* https://www.lentopallo.fi/koulutukset/lasten-lentopallovalmennus/ — **Course explicitly references applying ecological-dynamics principles** — key evidence for the "internal tension" framing (§2).
- **Suomen Lentopalloliitto.** Coach-education renewal 2026. https://www.lentopallo.fi/uutiset/lentopallon-valmentajakoulutus-kohdennetumman-ja-pitkajanteisemman-osaamisen-kehittymisen-tukena/ — Notes coach development is explicitly "non-linear."
- **Suomen Lentopalloliitto.** *Valmennuslinjaukset / lajianalyysi.* https://www.lentopallo.fi/valmennus/valmennuslinjaukset/ — Sport analysis + physical-training guidelines.
- **Koponen, S. (2019).** *Naisten lentopallon vastaanottohyökkäysvaiheen teknis-taktinen lajianalyysi* [Bachelor's thesis]. Liikuntatieteellinen tiedekunta, Jyväskylän yliopisto. https://jyx.jyu.fi/handle/123456789/66483 `[thesis]` — Finnish women's volleyball technical-tactical match analysis. (Companion working paper: Koponen, S. (2019). *Naisten lentopallon lajianalyysi ja valmennuksen ohjelmointi.*)

---

2. Volleyball-specific CLA / representative design

- **Moy, B., Renshaw, I., & Gorman, A. D. (2024).** Applying the constraints-led approach

to facilitate exploratory learning of the volleyball serve. *Journal of Sports Sciences, 42*(6), 511–518. https://doi.org/10.1080/02640414.2024.2345415 `[peer-reviewed]` — A 2-serve game (vs regulation single serve) produced faster, more complex first serves (60.3 vs 55.6 km/h; 44.5% vs 25.2% jump-topspin) by cutting fear; the value is the *exploration unlocked* (accuracy did not improve). Basis for design-guide Case A.

- **Caldeira, P., Paulo, A., Veloso, A., Infante, J., Davids, K., & Araújo, D. (2023).** How functional movement variability facilitates successful skill adaptation during the volleyball attack. *International Journal of Sports Science & Coaching.* https://doi.org/10.1177/17479541231209400 `[check: DOI/issue] [peer-reviewed]` — Young male hitters trained CLA vs traditional/blocked for the **spike against an opposing block**; after 6 weeks the CLA group had a significantly higher % of successful attacks. (Note: this is *attack vs a block*, not "blocking.")
- **Pepe, G., Albano, D., & Vastola, R. (2021).** A constraints-led approach in volleyball. *Italian Journal of Health Education, Sport and Inclusive Didactics, 5* (2). https://doi.org/10.32043/gsd.v5i2.366 `[peer-reviewed]` — Applied/conceptual piece on CLA in volleyball.
- **Renshaw, I., Davids, K., Newcombe, D., & Roberts, W. (2019).** *The constraints-led approach: Principles for sports coaching and practice design.* Routledge. `[book]` — The canonical CLA practice-design text; "keep it together, don't decompose."
- **Trinsey, J.** *Constrain to break through.* Smarter Volley. https://smartervolley.substack.com/p/constrain-to-break-through `[grey]` — Practitioner constraint logic for volleyball.
- **Kessel, J. / USA Volleyball.** *Growing the Game Together* (blog-book, vol. 3). https://usavolleyball.org/wp-content/uploads/2020/12/blog-book-volume-3.pdf `[grey]` — Game-based, random/varied practice; "the game teaches the game."

---

3. Volleyball perception–action & expertise (use carefully — see research-notes §4)

- **Mann, D. T. Y., Williams, A. M., Ward, P., & Janelle, C. M. (2007).** Perceptual-cognitive expertise in sport: A meta-analysis. *Journal of Sport and Exercise Psychology, 29*(4), 457–478. https://doi.org/10.1123/jsep.29.4.457 `[review]` — The anchor: experts read cues faster/more accurately with fewer, longer fixations + longer quiet eye; crucially, expertise effects are **largest under in-situ, representative conditions** and shrink with decoupled video tests — the empirical basis for §4's "use carefully."
- **Piras, A., Lobietti, R., & Squatrito, S. (2014).** Response time, visual search strategy, and anticipatory skills in volleyball players. *Journal of Ophthalmology, 2014,* 189268. https://doi.org/10.1155/2014/189268 `[peer-reviewed]` — Experts predict the setter's direction with fewer, longer fixations.
- **Fortin-Guichard, D., Laflamme, V., Julien, A.-S., Trottier, C., & Grondin, S. (2020).** Decision-making and dynamics of eye movements in volleyball experts. *Scientific Reports, 10,* 17288. https://doi.org/10.1038/s41598-020-74487-x `[peer-reviewed]` — Setters vs other experts vs controls; the opponent's upper body is the key cue right before contact.
- **Kanatschnig, T., Korda, Ž., Schrapf, N., Leitner, L., Anzengruber, C., Lappi, O., Körner, C., Tilp, M., & Kober, S. E. (2025).** Expertise-driven temporal gaze dynamics during anticipation in volleyball. *PLoS One, 20*(10), e0334702. https://doi.org/10.1371/journal.pone.0334702 `[peer-reviewed]` — Expert/amateur gaze differences appear in the *later* (rally) phase, not pre-serve — temporal dynamics matter. (Note: 2025, not 2024.)
- **In-situ representative-design study — backcourt defence.** *Journal of Sports Science & Medicine, 11*(2), 339–… `[check: authors/year – likely linked to Caldeira/Araújo group] [peer-reviewed]` — In-situ 6v6 defence shows skilled players use more exploratory search of functional spaces.
- **On-court vs off-court gaze study.** PMC10968438. `[check: full citation] [peer-reviewed]` — On-court vs video gaze behaviour differs (caution about decontextualised perception tests).

---

4. Empirical CLA / motor-learning comparison studies (read for §7)

- **Gray, R.** *Comparative training studies of traditional prescriptive coaching and "ecological" approaches* [running log]. https://perceptionaction.com/comparative/ `[grey]` — ~21 head-to-head studies; the distilled pattern is in research-notes §2.
- **Gray, R. (2018).** Comparing the constraints-led approach, differential learning and prescriptive instruction for training opposite-field hitting in baseball. *Human Movement Science, 61,* 159–174. https://doi.org/10.1016/j.humov.2018.07.011 `[check: vol/pages] [peer-reviewed]` — CLA improved both movement coordination *and* affordance perception; beat DL and prescriptive instruction.
- **Deuker, A., Braunstein, B., Chow, J. Y., Fichtl, M., Kim, H., Körner, S., & Rein, R. (2023).** "Train as you play": Improving effectiveness of training in youth soccer players.

International Journal of Sports Science & Coaching. https://doi.org/10.1177/17479541231172702 `[peer-reviewed]` — Decomposed passing drills vs small-sided games in 10-y-olds: both improved, but the games group retained better — even though the test favoured the drill group. Direct "fundamentals-first" counter-evidence.

- **Moy, B., Renshaw, I., & Pavey, T. (2020).** Impact of the constraints-led approach on students' motor performance. *Journal of Physical Education and Sport, 20* (6, Art. 453), 3345–3353. https://doi.org/10.7752/jpes.2020.06453 `[peer-reviewed]` — One 40-min CLA lesson significantly improved hurdling (a complex coordination skill).
- **Chapman, S. J. (2015).** *Spin it to win it: A comparison of constraints-led versus traditional coaching approaches* [Masters by Research thesis]. Queensland University of Technology. https://eprints.qut.edu.au/90731/ `[thesis]` — Cricket spin: CLA best for younger/less-skilled bowlers; traditional better for older/more-skilled — the skill-level **moderation** caveat.
- **Yang, Q., Song, M., Chen, X., Li, M., & Wang, X. (2025).** The influence of linear and nonlinear pedagogy on motor skill performance: the moderating role of adaptability. *Frontiers in Psychology, 16,* 1540821. https://doi.org/10.3389/fpsyg.2025.1540821 `[peer-reviewed]` — Soccer passing: NLP > LP for immediate gain *and* retention; adaptability moderates NLP's effect (high-adaptability learners gain most).
- **Bromilow, L., Milne, N., Woods, C. T., Dowsett, C. K., & Keogh, J. W. L. (2025).** The effectiveness of linear and nonlinear pedagogical approaches in team-invasion ball sports: A systematic review. *Sports Medicine – Open, 11* (1), 90. https://doi.org/10.1186/s40798-025-00893-y `[review]` — 9 studies: NLP ≈ linear for most technical outcomes (favoured in 34%), NLP > linear for tactical outcomes (66%); calls for higher-quality, more representative research. **(Primary anchor for Claim 2 — see Appendix B.)**
- **Abad Robles, M. T., Collado-Mateo, D., Fernández-Espínola, C., Castillo Viera, E., & Giménez Fuentes-Guerra, F. J. (2020).** Effects of teaching games on decision making and skill execution: A systematic review and meta-analysis. *International Journal of Environmental Research and Public Health, 17*(2), 505. https://doi.org/10.3390/ijerph17020505 `[review]` — Tactical (game-based) interventions significantly improved **decision-making** (ES ≈ 0.89) but showed **no significant difference for skill execution** vs technical approaches; evidence quality low (GRADE), heterogeneity high.
- **Ortiz, M., Meroño, L., Morales-Belando, M. T., Vaquero-Cristóbal, R., & González-Gálvez, N. (2023).** Teaching Games for Understanding in game performance and psychosocial variables: Systematic review and meta-analysis of randomized control trials. *Children, 10*(3), 573. https://doi.org/10.3390/children10030573 `[review]` — 13 RCTs: large improvement in **decision-making** for TGfU vs technical (SMD ≈ 5.93, but $I^2 = 98\%$), **no difference in skill execution** ; PEDro-appraised.
- **Application of the TGfU model to improve decision-making in sport learning (2024).** *BMC Psychology.* https://doi.org/10.1186/s40359-024-02307-2 `[check: author list] [review]` — 25 studies (n ≈ 1692): TGfU improves decision-making in sport learning.
- **Chow, J. Y., Meerhoff, L. A., Choo, C. Z. Y., Button, C., & Tan, B. S.-J. (2023).** The effect of nonlinear pedagogy on the acquisition of game skills in a territorial game. *Frontiers in Psychology, 14,* 1077065. https://doi.org/10.3389/fpsyg.2023.1077065 `[peer-reviewed]` — NLP > LP for football acquisition, **but transfer to floorball was weaker than expected** and the authors conclude **both LP and NLP "have a role," context-dependent** — useful counter-nuance for Claim 2.

---

5. Motor-learning theory: practice structure, variability & attentional focus (the cognitive counterpoint + bridges)

Cited so the model engages the established motor-learning literature fairly, not as a strawman. Note: some of these (schema theory, challenge point) sit in the information-processing tradition the model declines as its base theory (§4) — but their prescriptions* (vary practice; scale difficulty to the learner; avoid internal-focus cueing) converge with the eco-D design moves, which strengthens the practical case.*

- **Schmidt, R. A. (1975).** A schema theory of discrete motor skill learning. *Psychological Review, 82*(4), 225–260. https://doi.org/10.1037/h0076770 `[peer-reviewed]` — The IP-era rationale for *variability of practice*; a different-theory cousin of eco-D variability (same prescription, different mechanism).
- **Shea, J. B., & Morgan, R. L. (1979).** Contextual interference effects on the acquisition, retention, and transfer of a motor skill. *Journal of Experimental Psychology: Human Learning and Memory, 5*(2), 179–187. https://doi.org/10.1037/0278-7393.5.2.179 `[peer-reviewed]` — The classic: random (high-interference) practice beats blocked for retention and transfer — the original empirical case against block-drilling.
- **Guadagnoli, M. A., & Lee, T. D. (2004).** Challenge point: A framework for conceptualizing the effects of various practice conditions in motor learning. *Journal of Motor Behavior, 36*(2), 212–224. https://doi.org/10.3200/jmbr.36.2.212-224 `[peer-`

reviewed] — Nominal vs *functional* task difficulty; learning is optimised at a challenge point scaled to skill level — the scaffold for "scale the constraint to the learner" and the skill-level moderation in §7.

- **Wulf, G. (2013).** Attentional focus and motor learning: a review of 15 years. *International Review of Sport and Exercise Psychology, 6* (1), 77–104. https://doi.org/10.1080/1750984X.2012.723728 [review] — External focus (on the effect) beats internal focus (on body parts). Backs the *lajilinjaukset*' external cues — but also indicts the step-by-step footwork scripting that sits beside them (§1).
- **Schöllhorn, W. I., Paschke, M., & Beckmann, H. (2006).** Differenzielles Training im Volleyball beim Erlernen von zwei Techniken. In K. Langolf & R. Roth (Eds.), *Volleyball 2005 – Beach-WM* (pp. 97–105). Czwalina. [check] — Early volleyball differential-learning study.
- **Apidogo, J. B., Burdack, J., & Schöllhorn, W. I. (2021).** Repetition without repetition or differential learning of multiple techniques in volleyball? *International Journal of Environmental Research and Public Health, 18*(19), 10499. https://doi.org/10.3390/ijerph181910499 [peer-reviewed] — In adolescent girls learning underhand pass, overhand pass and serve in parallel, **differential learning beat both repetitive and general-familiarisation** groups on every technique at retention. Volleyball-specific support for "no one ideal technique."
- **Sattelmayer, M., Elsig, S., Hilfiker, R., & Baer, G. (2016).** A systematic review and meta-analysis of selected motor learning principles in physiotherapy and medical education. *BMC Medical Education, 16,* 15. https://doi.org/10.1186/s12909-016-0538-z [review] — Across 15 RCTs: **whole practice showed some advantages over part practice**, and random > blocked, on post-acquisition tests — *but all studies were high risk of bias*. Domain is procedural/medical skills, so transfer to sport is indirect. Used in Appendix B as cross-domain support (and an honesty check) for Claim 1.
- **Nixon, J. E., & Locke, L. F. (1973).** Research on teaching physical education. In R. M. W. Travers (Ed.), *Second handbook of research on teaching* (pp. 1210–1242). Rand McNally. [book] [check: pages] — Classic review frequently cited for the claim that the comparative literature did **not** identify studies favouring part over whole training; old and narrative, included for provenance of the "whole-practice superiority" idea, not as current evidence. **Counterpoint to hold in mind:** the classic part–whole tradition (Naylor & Briggs, 1963) holds that **part practice can help for tasks high in complexity and low in part-interdependence** — i.e. "never decompose" is too strong (see Appendix B).

---

6. Ecological dynamics, CLA & nonlinear pedagogy — theory

- **Gibson, J. J. (1979).** *The ecological approach to visual perception.* Houghton Mifflin. [book] — Origin of *affordances* and direct perception — the foundation under the model's "skill = using information" definition (§7).
- **Newell, K. M. (1986).** Constraints on the development of coordination. In M. G. Wade & H. T. A. Whiting (Eds.), *Motor development in children: Aspects of coordination and control* (pp. 341–360). Martinus Nijhoff. [book] — The individual/task/environment constraints model (the model's design vocabulary).
- **Araújo, D., Davids, K., & Passos, P. (2007).** Ecological validity, representative design, and correspondence between experimental task constraints and behavioral setting. *Ecological Psychology, 19*(1), 69–78. https://doi.org/10.1080/10407410709336951 [peer-reviewed] — The precise (Brunswikian) definition of *representative design* the model uses for the indoor-representativeness caveat (§6a).
- **Davids, K., Button, C., & Bennett, S. J. (2008).** *Dynamics of skill acquisition: A constraints-led approach.* Human Kinetics. [book] — The foundational CLA textbook (1st ed.).
- **Chow, J. Y., Davids, K., Button, C., Shuttleworth, R., Renshaw, I., & Araújo, D. (2007).** The role of nonlinear pedagogy in physical education. *Review of Educational Research, 77*(3), 251–278. https://doi.org/10.3102/003465430305615 [peer-reviewed] — Foundational NLP statement: variability, self-organisation, emergent decision-making.
- **Renshaw, I., Chow, J. Y., Davids, K., & Hammond, J. (2010).** A constraints-led perspective to understanding skill acquisition and game play: A basis for integration of motor learning theory and physical education praxis? *Physical Education and Sport Pedagogy, 15*(2), 117–137. https://doi.org/10.1080/17408980902791586 [peer-reviewed] — CLA applied to game play / learning environments.
- **Pinder, R. A., Davids, K., Renshaw, I., & Araújo, D. (2011).** Representative learning design and functionality of research and practice in sport. *Journal of Sport and Exercise Psychology, 33*(1), 146–155. https://doi.org/10.1123/jsep.33.1.146 [peer-reviewed] — Representative design, action fidelity, perception–action coupling.
- **Chow, J. Y., Davids, K., Button, C., & Renshaw, I. (2016).** *Nonlinear pedagogy in skill acquisition: An introduction.* Routledge. [book] — Book-length NLP treatment.
- **Renshaw, I., & Chow, J. Y. (2019).** A constraint-led approach to sport and physical education pedagogy. *Physical Education and Sport Pedagogy, 24* (2), 103–116. https://doi.org/10.1080/17408989.2018.1552676 [check: vol/pages] [peer-reviewed] — Concise CLA-for-pedagogy statement; good single citation for coach-

education material.

- **Woods, C. T., McKeown, I., Rothwell, M., Araújo, D., Robertson, S., & Davids, K. (2020).** Sport practitioners as sport ecology designers: How ecological dynamics has progressively changed perceptions of skill "acquisition" in the sporting habitat. *Frontiers in Psychology, 11,* 654. https://doi.org/10.3389/fpsyg.2020.00654 `[peer-reviewed]` — Reframes the coach as designer of an athlete–environment relationship.
- **Chow, J. Y., Button, C., Lee, M. C. Y., Morris, C., & Shuttleworth, R. (2023).** Advice from "pracademics" of how to apply ecological dynamics theory to practice design. *Frontiers in Sports and Active Living, 5,* 1192332. https://doi.org/10.3389/fspor.2023.1192332 `[peer-reviewed]` — On the jargon/abstraction barrier and bridging theory to the court (relevant to §0 "show, don't just tell"). *(Corrects an earlier mis-attribution to "Tan et al.")*
- **Kennedy, … & O'Brien, … (2024).** Adding texture to the art of constraints-led coaching: A request for more research-informed guidelines. *Sports Coaching Review.* https://doi.org/10.1080/21640629.2024.2395135 `[check: initials] [peer-reviewed]` — Coaches lack research-informed guidance for choosing/manipulating constraints; much of CLA is still craft.
- **Gray, R. (2021).** *How we learn to move: A revolution in the way we coach and practice sports skills.* Perception & Action Consulting. `[book]` — Accessible eco-D synthesis; argues IP and ecological dynamics are not coherently blendable.
- **Gray, R. (2025).** *The advanced ecological approach to skill development.* Perception & Action. `[book]` — Esp. Ch. 7, "Adjusting the 'dials'… the better alternative to a 'fundamentals first' approach" (task simplification vs decomposition). Ch. 7 also makes the **assessment-bias** point used in Appendix B: many comparative studies measured "technique" with isolated/closed tests that, by specificity-of-practice, favour the traditional group — so eco-D reaching parity on those tests *understates* its value.
- **Gray, R.** *Perception & Action* [podcast]. e.g. Ep. 383, "Why I switched sides to the ecological approach." https://perceptionaction.com/383-2/ `[grey]` — Source for the IP-vs-eco-D coherence argument.
- **Button, C., Seifert, L., Chow, J. Y., Davids, K., & Araújo, D. (2020).** *Dynamics of skill acquisition: An ecological dynamics approach* (2nd ed.). Human Kinetics. `[book]` — Standard eco-D textbook (the updated edition of Davids, Button & Bennett 2008).

---

7. Technique vs skill — definitions (research-notes §8)

- **Bernstein, N. A. (1967).** *The co-ordination and regulation of movements.* Pergamon Press. `[book]` — Origin of "repetition without repetition"; degrees-of-freedom problem.
- **Bernstein, N. A. (1996).** *On dexterity and its development* (M. L. Latash & M. T. Turvey, Eds.; M. L. Latash, Trans.). Lawrence Erlbaum. `[book]` — Dexterity = "the ability to find a motor solution for any external situation… correctly, quickly, rationally, and resourcefully"; "create a perfect key for any emerging lock."
- **Newell, K. M. (1985).** Coordination, control and skill. In D. Goodman, R. B. Wilberg, & I. M. Franks (Eds.), *Differing perspectives in motor learning, memory, and control* (pp. 295–317). North-Holland. `[book]` — Coordination → control → skill; learning as search of the perceptual-motor workspace.
- **Araújo, D., & Davids, K. (2011).** What exactly is acquired during skill acquisition? *Journal of Consciousness Studies, 18* (3–4), 7–23. `[peer-reviewed]` — Skill as "the emergence of an adaptive, functional relationship between an organism and its environment"; "organismic asymmetry"; "skill attunement."
- **Woods, C. T., & Davids, K. (2021).** Enskilment: An ecological-anthropological worldview of skill, learning and education in sport. *Sports Medicine – Open, 7,* 33. https://doi.org/10.1186/s40798-021-00326-6 `[check: full author list] [peer-reviewed]` — Skill as progressive attunement / education of attention.
- **Guthrie, E. R. (1952).** *The psychology of learning* (Rev. ed.). Harper & Row. `[book]` — Classic skill definition (max certainty, min energy/time).
- **Knapp, B. (1963).** *Skill in sport: The attainment of proficiency.* Routledge & Kegan Paul. `[book]` — "The learned ability to bring about predetermined results with maximum certainty…" — the coaching-tradition (distinction A) lineage.
- **Sarama, A. (2024).** *Transforming basketball: Changing how we think about basketball performance.* https://transformingbball.com/ `[grey]` — "Skills are not actions or techniques… they are the relationship between the player and their environment." Source of the Swedish/Finnish basketball-model thread.
- *Practitioner definitions (distinction A vs B):* BrianMac "Skill, technique and ability" (https://brianmac.co.uk/skills.htm); Soccer Coach Weekly; Skilled Athleticism (technique = a coordination pattern; skill = adapting it to functional outcomes). `[grey]`

---

8. Game-based pedagogies (adjacent traditions coaches already know)

Not eco-D, but convergent game-first families the model can position against/alongside —

useful for adoption and for "show, don't just tell."

- **Bunker, D., & Thorpe, R. (1982).** A model for the teaching of games in secondary schools. *Bulletin of Physical Education, 18*(1), 5–8. `[peer-reviewed]` — The original Teaching Games for Understanding (TGfU) model; game-then-technique.
- **Launder, A. G. (2001).** *Play practice: The games approach to teaching and coaching sports.* Human Kinetics. `[book]` — "Play Practice"; shaping/focusing/enhancing play. (2nd ed.: Launder & Piltz, 2013.)
- **Light, R. L. (2013).** *Game Sense: Pedagogy for performance, participation and enjoyment.* Routledge. `[book]` — Game Sense (the Australian games-based coaching tradition).
- **Gil-Arias, A., Diloy-Peña, S., Sevil-Serrano, J., García-González, L., & Abós, Á. (2021).** A hybrid TGfU/SE volleyball teaching unit for enhancing motivation in physical education: A mixed-method approach. *International Journal of Environmental Research and Public Health, 18*(1), 110. https://doi.org/10.3390/ijerph18010110 `[check: full author list] [peer-reviewed]` — SDT-grounded; a volleyball game-based unit improved need-satisfaction and intrinsic motivation, with a *larger* effect for girls — relevant to participation/retention and the whole-person layer.

---

9. Implicit learning, deliberate play & equipment scaling (the ball-taming fieldstone)

- **Côté, J., & Hay, J. (2002).** Children's involvement in sport: A developmental perspective. In J. M. Silva & D. Stevens (Eds.), *Psychological foundations of sport* (pp. 484–502). Allyn & Bacon. `[book]` — Deliberate play vs deliberate practice; the DMSP.
- **Wall, M., & Côté, J. (2007).** Developmental activities that lead to dropout and investment in sport. *Physical Education and Sport Pedagogy, 12* (1), 77–87. https://doi.org/10.1080/17408980601060358 `[peer-reviewed]` — Early focus should be play/fun, not intense routine training.
- **Masters, R. S. W. (1992).** Knowledge, knerves and know-how: The role of explicit versus implicit knowledge in the breakdown of a complex motor skill under pressure. *British Journal of Psychology, 83* (3), 343–358. https://doi.org/10.1111/j.2044-8295.1992.tb02446.x `[peer-reviewed]` — Foundational implicit-motor-learning / reinvestment work.
- **Capio, C. M., Poolton, J. M., Sit, C. H. P., Eguia, K. F., & Masters, R. S. W. (2012).** The possible benefits of reduced errors in the motor skills acquisition of children. *Sports Medicine, Arthroscopy, Rehabilitation, Therapy & Technology, 4,* 1. PMC3275454. `[check: author order] [peer-reviewed]` — Implicit/error-reduced learning suits children's maturing working memory.
- **Kal, E., Prosée, R., Winters, M., & van der Kamp, J. (2018).** Does implicit motor learning lead to greater automatization of motor skills compared to explicit motor learning? A systematic review. *PLoS One, 13*(9), e0203591. PMC6124806. https://doi.org/10.1371/journal.pone.0203591 `[check: vol/article] [review]` — Some evidence implicit learning resists pressure/fatigue better.
- **Buszard, T., Farrow, D., Reid, M., & Masters, R. S. W. (2014).** Scaling sporting equipment for children promotes implicit processes during performance. *Consciousness and Cognition, 30,* 247–255. https://doi.org/10.1016/j.concog.2014.07.004 `[check: vol/pages] [peer-reviewed]` — Scaled/lighter equipment reduces conscious processing; volleyball parallel for ball/net/court scaling.

---

10. Talent development, LTAD frameworks & physical literacy (incl. the critique)

- **Volleyball Canada (2024).** *Long-Term Development 3.0.* https://volleyball.ca/uploads/About/LTAD/LTD3.0_240111.pdf `[org]` — Stage structure (Active Start → … → Win/Active for Life) used to scaffold §10.
- **Volleyball Ireland (2020).** *Long-Term Player Development model overview.* https://www.volleyballireland.com/wp-content/uploads/2020/12/LTPD-Model-Overview-Guide.pdf `[org]` — Comparator LTPD.
- **Sport for Life (2019).** *Long-Term Development in Sport and Physical Activity 2.1.* https://sportforlife.ca/wp-content/uploads/2017/04/LTAD-2.1-EN_web.pdf `[org]` — The Sport for Life stage rectangle / late-specialisation classification.
- **Wormhoudt, R., Savelsbergh, G. J. P., Teunissen, J. W., & Davids, K. (2018).** *The Athletic Skills Model: Optimizing talent development through movement education.* Routledge. https://doi.org/10.4324/9781315201474 `[book]` — An eco-D-aligned *alternative* to standard LTAD: broad movement education / "physical intelligence" before specialisation. Directly relevant comparator for §10.
- **Bailey, R., & Collins, D. (2013).** The standard model of talent development and its discontents. *Kinesiology Review, 2*(4), 248–259. https://doi.org/10.1123/krj.2.4.248 `[peer-reviewed]` — The critique: the pyramid/early-selection "standard model" is

unsupported on both process and outcome — keeps the model honest about leaning on LTAD scaffolding.
- **Ford, P. R., Ward, P., Hodges, N. J., & Williams, A. M. (2009).** The role of deliberate practice and play in career progression in sport: the early engagement hypothesis. *High Ability Studies, 20*(1), 65–75. https://doi.org/10.1080/13598130902860721 `[peer-reviewed]` — Early diversified *play* (not early specialisation) characterises those who progress — supports the play-rich early stages.
- **Whitehead, M. (Ed.). (2010).** *Physical literacy: Throughout the lifecourse.* Routledge. `[book]` — The physical-literacy construct invoked in the Active Start / FUNdamentals stage (§10).

---

11. Finnish-language pedagogy, coaching competence & athlete pathway

- **Hämäläinen, K. (toim.). (2012; updated 2016).** *Suomalaisen valmennusosaamisen malli* / *Valmennusosaamisen käsikirja.* Suomen Olympiakomitea. `[org]` — **The Finnish coaching-competence framework** the federation's coach education is built on (urheiluosaaminen → ihmissuhdetaidot → itsensä kehittämisen taidot, within a *toimintaympäristö*; "what the coach can influence in the athlete"). Built on Côté & Gilbert (2009). Adoption-critical: the model should speak this vocabulary.
- **Jaakkola, T. (2017).** *Liikuntataitojen oppiminen ja taitoharjoittelu.* `[check: year/publisher]` `[book]` — Finnish skill-learning reference; established terminology.
- **Kalaja, S., & Laakso, T. (2026).** *Taitoharjoittelu.* VK-Kustannus. `[check: forthcoming]` `[book]` — First comprehensive Finnish skill-training reference (per research-notes §5).
- **Kalaja, S.** Taitoharjoittelun uudet (vanhat) suunnat. `[grey]` `[check: venue]` — NLP in Finnish (*rajoitteet, representatiivinen oppimisen suunnittelu, "näpit irti" - valmentaminen*).
- **Konttinen, J. (2016).** *Epälineaarinen pedagogiikka ja differentiaalioppiminen.* Suomen Suunnistusliitto. https://www.suunnistusliitto.fi/system/wp-content/uploads/2016/01/2016_EpalineaarinenPedagogiikkaDifferentiaalioppiminen_KonttinenJuuso.pdf `[grey]` — Finnish-language NLP/differential-learning primer; shows the inconsistent *epälineaarinen/ei-lineaarinen/non-lineaarinen* usage.
- **Suomen Olympiakomitea.** *Urheilijan polku / "Kasva urheilijaksi"* (lapsuus-/valinta-/huippuvaihe). https://www.olympiakomitea.fi/ `[org]` — Finnish phase model for §10 localisation.
- **Nuori Suomi / asiantuntijatyöryhmä (2012).** *Lasten urheilun asiantuntijatyö.* `[org]` `[check]` — Childhood-phase "punainen lanka" (Côté DMSP basis).

---

12. Coaching effectiveness, motivation & the whole-person layer (MI + SDT)

- **Côté, J., & Gilbert, W. (2009).** An integrative definition of coaching effectiveness and expertise. *International Journal of Sports Science & Coaching, 4* (3), 307–323. https://doi.org/10.1260/174795409789623892 `[peer-reviewed]` — Coaching effectiveness = professional + interpersonal + intrapersonal knowledge applied to athletes' competence, confidence, connection, character. The definition under the Finnish *valmennusosaaminen* model (§11).
- **Deci, E. L., & Ryan, R. M. (2000).** The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry, 11*(4), 227–268. https://doi.org/10.1207/S15327965PLI1104_01 `[peer-reviewed]` — Self-Determination Theory: autonomy, competence, relatedness — the foundation under the whole-person/motivation layer.
- **Ryan, R. M., & Deci, E. L. (2017).** *Self-determination theory: Basic psychological needs in motivation, development, and wellness.* Guilford Press. `[book]` — The comprehensive SDT statement.
- **Miller, W. R., & Rollnick, S. (2013).** *Motivational interviewing: Helping people change* (3rd ed.). Guilford Press. `[book]` — Canonical MI text (spirit of MI; OARS; processes).
- **Rollnick, S., Fader, J., Breckon, J., & Moyers, T. B. (2019).** *Coaching athletes to be their best: Motivational interviewing in sports.* Guilford Press. `[book]` — Sport-specific MI.
- **Markland, D., Ryan, R. M., Tobin, V. J., & Rollnick, S. (2005).** Motivational interviewing and self-determination theory. *Journal of Social and Clinical Psychology, 24*(6), 811–831. https://doi.org/10.1521/jscp.2005.24.6.811 `[peer-reviewed]` — The MI ↔ SDT link.
- **Mack, R., Breckon, J., O'Halloran, P., & Butt, J. (2017).** Enhancing athlete engagement in sport psychology interventions using motivational interviewing: A case study. *The Sport Psychologist, 31*(3), 1–… https://doi.org/10.1123/tsp.2016-0125 `[check: title/authors]` `[peer-reviewed]` — Applied MI in sport psychology + the limited sport evidence base.
- **Meta-analytic context (physical activity):** e.g. a *BMJ* (2024) review of MI-containing

interventions and physical activity. `[check: full citation] [review]`

---

13. Internal / cross-reference documents (this repo)

- `swedish-basketball-development-model.md` / `.pdf` — Swedish basketball federation model brief (eco-D/CLA inspiration).
- `finnish-basketball-development-model.md` / `.pdf` — Finnish basketball model brief (humanistic core, konseptiajattelu).
- `fieldstones.md` — ball-taming, messy-practice, change-the-game-not-the-player, technique-vs-skill stones.
- `finnish-volleyball-model-research-notes.md` — the research log (full distillations and quotes behind these sources).

Appendix A — Documented search strategy

*This appendix records **how** the literature behind the model was sought, so a reader can judge coverage and reproduce it. It is a documented **scoping-style** strategy, not a completed systematic review: see the honesty note at the end.*

A.1 Review question (PICO-style)

For the two claims the model rests on (operationalised in Appendix B):

Element	Scope
Population	Learners of sport motor skills — emphasis on youth/novice and net/invasion team-sport players; volleyball where available, transferable team-sport evidence otherwise.
Intervention	Ecological / nonlinear / constraints-led / game-based practice (CLA, NLP, TGfU, Game Sense, differential learning, representative design, **task simplification**).
Comparison	"Traditional" / linear / technique-first / direct-instruction / **task-decomposition** ("fundamentals first", isolated/blocked drills).
Outcomes	Skill/technique execution; tactical & decision-making performance; retention; transfer; (secondarily) motivation/enjoyment.
Study types	Systematic reviews & meta-analyses (priority), RCTs and controlled intervention studies, plus key theory and authoritative grey literature (books, federation documents).

A.2 Information sources

- **Indexed databases / publisher platforms searched (via web interface):** PubMed, Scopus-indexed and Web of Science–indexed records surfaced through publisher pages (MDPI, Springer/BMC, Frontiers, Taylor & Francis), SPORTDiscus-indexed records where openly visible, and Google Scholar–style web search.
- **Hand/snowball searching:** reference lists of the included reviews (Bromilow 2025; Abad Robles 2020; Ortiz 2023; Sattelmayer 2016); citation chains around key authors (Davids, Renshaw, Chow, Gray, Schöllhorn).
- **Authoritative grey literature:** Rob Gray's *Perception & Action* book and podcast log (the running comparative tally of CLA-vs-traditional studies), Finnish Volleyball Federation *lajilinjaukset* and Junnulentis/PowerCup youth rules.

A.3 Search strings (concept blocks, combined with AND)

Block 1 — **approach:** `("ecological dynamics" OR "constraints-led" OR "constraint-led" OR "nonlinear pedagogy" OR "non-linear pedagogy" OR`

`"differential learning" OR "teaching games for understanding" OR TGfU OR "game-based approach" OR "game sense" OR "representative design" OR "task simplification")`

Block 2 — **comparator:** `("traditional" OR linear OR "technique-first" OR "direct instruction" OR "part practice" OR "part-task" OR "task decomposition" OR "blocked practice" OR "fundamentals first" OR "drill*")`

Block 3 — **outcome:** `("skill acquisition" OR "motor learning" OR "skill execution" OR technique OR "decision making" OR tactical OR retention OR transfer)`

Block 4 — **domain (optional narrowing):** `(volleyball OR "net sport" OR "invasion game" OR "team sport" OR novice OR youth OR children OR beginner)`

Block 5 — **design (for the review tier):** `("systematic review" OR meta-analysis OR "randomi*ed controlled trial" OR RCT OR "controlled trial")`

A.4 Inclusion / exclusion criteria

Include if: (a) compares an ecological/nonlinear/game-based approach against a traditional/linear/decomposed approach, **or** is a systematic review/meta-analysis of such comparisons; (b) reports a skill, tactical, retention or transfer outcome; (c) human participants learning a sport/motor skill; (d) English- or Finnish-language; (e) any year (no date cap — foundational theory deliberately retained).

Exclude if: purely theoretical/opinion with no data **and** not a foundational/definitional source the model cites; non-sport clinical-only populations (kept only where used explicitly as cross-domain evidence, e.g. Sattelmayer 2016, and flagged as such); coaching-perception/qualitative-only studies (logged separately, not used as efficacy evidence).

A.5 Screening & extraction

Titles/abstracts screened against A.4; full text retrieved for likely-eligible records. From each included source extracted: design, n, age/level, sport, comparison, primary outcome(s) and **direction of effect**. Reviews were preferred as the top evidence tier; primary studies were used to add domain specificity (esp. volleyball) and to surface **counter-evidence and null results**.

A.6 Honesty note (what this is *not*)

This was executed by a single reviewer through web/publisher interfaces, **not** a controlled multi-database export with two independent screeners, a PRISMA flow count, or formal risk-of-bias scoring beyond what the included reviews already report. Record counts in Appendix B are therefore **approximate** and the synthesis is **narrative**. Treat this as a transparent, reproducible *scoping* step toward — not a substitute for — a registered PRISMA(-ScR) review.

Appendix B — Scoping pass on the two load-bearing claims

The whole model leans on two empirical claims. Here each is stated precisely, mapped to the evidence found via Appendix A, and given an honest verdict — including where the strong version of the claim overreaches.

Claim 1 — "Make it easier by simplifying the whole task, not by decomposing it into drilled fundamentals."

(i.e. for beginners, scaling down the whole game beats breaking it into isolated parts; "fundamentals first" has no general advantage.)

Source	Type / design	Domain	Finding & direction
Deuker et al. (2023)	Intervention	Soccer (youth)	Small-sided games ≥ decomposed passing drills, **better retention** → favours simplification
Apidogo et al. (2021)	RCT-style	**Volleyball** (adolescent)	Differential (variable, whole-skill) > repetitive & general on all techniques at retention → favours variability over fixed-template drilling
Caldeira et al. (2023)	Intervention	**Volleyball** (spike vs block)	CLA practice → higher % successful attacks than traditional/blocked → favours coupled, representative practice
Gray (2025, Ch. 7) + P&A log	Book / running tally	Multi-sport	Synthesises ~20+ comparisons as "simplification not decomposition" → favours simplification
Sattelmayer et al. (2016)	Review (15 RCTs)	Procedural/medical	**Whole > part** on post-acquisition (but high risk of bias) → cross-domain support
Nixon & Locke (1973); Davids et al. (1999)	Classic review; volleyball	Multi / volleyball serve	No studies favouring part>whole; decoupled toss-only practice distorts trajectory → favours whole
Counter-nuance: Naylor & Briggs (1963); Human Kinetics part–whole guidance	Theory / textbook	Generic motor skills	**Part practice can help** for tasks **high in complexity, low in part-interdependence** → "never decompose" is too strong

Verdict (Claim 1): Well-supported in its *defaults* form, overreaching in its *absolute* form. For the interdependent, perception-coupled skills that make up a volleyball rally (pass–set–attack under a live ball), the evidence consistently favours **task simplification** (scale the whole game: lighter/slower ball, lower net, smaller court, fewer players) over **decomposition** (isolated, decoupled drills). The honest boundary — which the model already states (§7) — is that **isolated work has a narrow, bounded role** for components that are genuinely complex *and* separable (e.g. the serve toss), so the claim should read *"simplification is the better default,"* not *"decomposition is always wrong."* Evidence quality is moderate (several small intervention studies; the directly on-point reviews are cross-domain or running tallies rather than a sport-specific meta-analysis).

Claim 2 — "Nonlinear/ecological (CLA/NLP/game-based) practice is at least as good as linear/technique-first instruction, and better for tactics/decisions/transfer."

Source	Type / design	Domain	Finding & direction
Bromilow et al. (2025)	Systematic review (9 studies)	Team-invasion	NLP ≈ linear on most **technical** outcomes; NLP > **linear on tactical** (66%) → supports
Abad Robles et al. (2020)	Meta-analysis (6–7 studies)	Games/PE	Tactical: **decision-making** ES ≈ 0.89 (favours game-based); **skill execution: no sig. difference** → parity on technique, edge on decisions
Ortiz et al. (2023)	Meta-analysis (13 RCTs)	Games/PE	TGfU: large **decision-making** gain (I^2 = 98%); **skill execution: no difference** → same pattern
BMC Psychology (2024)	Meta-analysis (25 studies)	Sport learning	TGfU improves **decision-making** → supports
Yang et al. (2025)	Intervention	Soccer	NLP > LP acquisition & retention, **moderated by adaptability** → supports, with moderator
Chow et al. (2023)	Intervention	Football→floorball	NLP > LP football acquisition, **but transfer weaker than expected; "both have a role"** → mixed/counter
Moderators / counter: Chapman et al. (2015); Yang et al. (2025)	Primary	Various	Effects **moderated by skill level / adaptability** — not uniform superiority

Verdict (Claim 2): Supported, but must be stated precisely. The accurate claim is two-part: (a) on **technical/skill execution**, nonlinear ≈ linear — **parity, not superiority** (every meta-analysis found *no significant difference*); (b) on **tactical/decision-making** outcomes, nonlinear is **consistently favoured**. So the model should *not* claim NLP makes players technically better than drilling does — it claims NLP gets you **the same technique plus better game-reading, motivation, and (probably) transfer**, which is exactly why it's the better bet for a *game*. The caveats are real and should travel with the claim: small study counts, **low GRADE/PEDro quality and very high heterogeneity** , assessment instruments often not representative, and **moderation by skill level/adaptability** (the already-skilled and older may need less of it). Chow et al. (2023) is a useful humility check: transfer is not automatic.

One asymmetry strengthens the claim, though (Gray, 2025, Ch. 7; and Bromilow et al. 2025's own caveat). In most of these comparisons "technique" was tested in an **isolated, decontextualised** way — i.e. on the *traditional* group's home turf, since by specificity-of-practice a closed test favours whoever drilled closed. That the eco-D/NLP group nonetheless reached **parity even on that biased measure** means parity *understates* its value; a *representative* assessment (in a real rally) would likely have widened the gap. The cleanest example is **Deuker et al. (2023)**, where the outcome measure *was* the decomposed "fundamentals" test and the small-sided-games group still matched it **and retained better**. Bromilow et al. (2025) flag the same blind spot from the other side — calling for research that examines "how [results] may be implicated by the representativeness of the assessment instruments."

B.1 What this changes in the model

- **Tighten the wording of Claim 2** wherever the drafts imply NLP produces *better technique*: the defensible line is **"equal technique, better decisions/transfer."**
- **Keep Claim 1's bounded form** (simplification as default; decomposition narrow and bounded) — the scoping pass confirms the absolute "decomposition is wrong" version is unsupported by the classic part–whole literature.

- **Foreground the quality caveat** once, plainly: the direction of evidence favours the model, but the base is thin and low-quality, so the model is presented as **best-available-evidence + coherent theory**, not settled science. This is also the conclusion of the strongest review we have (Bromilow et al., 2025).

Maintenance: when a source is added to any draft, add it here first (with a type tag), then cite. Resolve `[check: …]` flags before any federation-facing release.